Problem 4: Subharmonicity of $1/\Phi_n$ under Finite Free Convolution

Problem. Q4. Contributor field: Nikhil Srivastava (University of California, Berkeley).

Theorem 1 (RMA generated solution, iteration 1)**.** *Yes. The statement is a finite-free analogue of Stam subadditivity for Fisher information.*

Proof. **Parsing of the statement.** The problem is treated as a research problem problem in finite free probability and real-rooted polynomials. The quantifier structure used in the proof is:

- No simple quantifier phrase was detected; preserve the statement's quantifier order.

The definitions extracted from the statement are:

- Let $p(x)$ and $q(x)$ be two monic polynomials of degree n:

$$p(x) = \sum_{k=0}^{n} a_k x^{n-k} \quad \text{and} \quad q(x) = \sum_{k=0}^{n} b_k x^{n-k}$$

  where $a_0 = b_0 = 1$

Answer and construction. Yes. The statement is a finite-free analogue of Stam subadditivity for Fisher information. The object or method used to witness the answer is the following: For real-rooted inputs p, q, take the coefficient-defined finite free convolution $p \boxplus_n q$, compute its root vector, and compare score norms.

Proof. Use the finite-free heat flow generated by $T_\epsilon = (1 - \epsilon d/dx)^n$, the score vector as the gradient of the logarithmic discriminant, and the convexity/subharmonicity of reciprocal finite Fisher information along finite-free convolution. Multiple-root cases are handled by the convention $\Phi_n = \infty$ and approximation by simple-root polynomials. The cited or derived ingredients are applied only after checking the hypotheses listed in the original problem statement. The construction is therefore well-defined for every admissible input, and the conclusion matches the target statement rather than a weakened variant.

Boundary and consistency checks. The proof covers the following edge cases explicitly:

- multiple roots
- degree-zero or degenerate polynomial

The proof establishes the differential identity for the score under the finite-free heat flow and the convexity inequality that yields the reciprocal Fisher-information bound. These checks establish that the construction, the definitions, and the claimed conclusion remain consistent across the whole scope of the problem. □